

Eric Waters

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Software Engineer with a M.S. in Big Data Systems and 3+ years of full-stack experience at Intel and Dell. Proven expertise in building robust APIs, data pipelines, AI/ML integrations, responsive UIs, and cloud/DevOps solutions.

Professional Experience

Software Application Development Engineer, Intel Corporation – Chandler, AZ Aug 2023 – Present

Batch Data Processing Platform

- Designed and built a standardized batch processing framework that **unifies disparate manufacturing ETL implementations** using Airflow, Docker, Kubernetes, GitHub Actions, and ELK.
- Created **reusable platform tooling** for ETL development, logging, CI/CD, and DAG creation, enabling teams to rapidly build and deploy pipelines with consistent infrastructure, performance, and quality.

Generative AI Web Platform

- Developed a **suite of 5 enterprise GenAI applications** using Angular and Python (Sanic), including a legal contract auditing tool estimated to save \$30M annually and recognized with a Divisional Recognition Award.
- Designed and developed a **no-code AI application studio** using React, Next.js, and Tailwind CSS, enabling non-technical teams to create custom AI workflows through an intuitive drag-and-drop interface.
- Transformed the platform UI/UX into a **modern, visually polished experience** and built **reusable frontend components** that improved consistency and accelerated feature delivery across applications.

Autonomous Excursion Response API

- Led consolidation of two large-scale FastAPI services** into a unified API, standardizing authentication, CI/CD workflows, Kubernetes deployments, and logging, reducing long-term complexity and maintenance overhead.

Factory ML Analysis API

- Engineered an asynchronous **FastAPI service for internal ML algorithms**, enabling external applications to execute long-running analytics jobs and retrieve ranked correlation results from factory-scale datasets.
- Designed a blue-green deployment architecture on Windows VMs and automated CI/CD with GitHub Actions, enabling **seamless failover and zero-downtime deployments**.

AI Enterprise Search Engine

- Administered an AI-powered enterprise search platform, developing Python ETL pipelines and managing 4 VMware servers to **index millions of manufacturing documents** and unify fragmented knowledge sources.

Graduate Software Engineering Intern, Dell Technologies – Round Rock, TX Jun 2022 – Aug 2022

- Implemented **OAuth 2.0 SSO and fine-grained authorization controls** across multiple enterprise web applications and REST APIs using Spring Boot and Angular, centralizing authentication, strengthening security and compliance, and streamlining user access.

Undergraduate Software Engineering Intern, Dell Technologies – Round Rock, TX May 2021 – Aug 2021

- Built a **custom Kanban board application** using Spring, improving agile workflow visibility and collaboration across engineering teams.

Technical Skills

Software Development: Python, Java, C, TypeScript, React, Next.js, Angular, FastAPI, Spring, SQL, OAuth 2.0

Data Engineering: Airflow, Pandas, Kafka, Spark, PostgreSQL, MySQL, SQL Server

Cloud & DevOps: Docker, Kubernetes, GitHub Actions, CI/CD, Rancher, Harbor, ELK Stack

Education

Arizona State University – Tempe, AZ

M.S. Computer Science (Big Data Systems), 4.0 GPA

May 2023

B.S. Computer Science, Summa Cum Laude, Barrett Honors College, 4.0 GPA

May 2022